

As shown in Fig. 2, these are:

1. Cones that sense colours
2. Rods that sense dim lights
3. Ganglion cells that sense blue light

When the eye sees an image, SCN receives information about incoming light because it is located just above the optic nerves. SCN relays information from the eyes to the brain. So, when the eyes see less at night, SCN tells the brain to produce melatonin, which makes us sleepy.

This melatonin production is in the pineal gland and its production varies with the time of day and also the colour of the light. The ganglion cells (which are sensitive to blue light) send signals to the brain and regulate the hormone production.

When light is switched on suddenly in the middle of the night, it leads to 'melatonin block,' which stops melatonin production, leading to fatigue during the day. If the body does not produce enough melatonin, it results in blood pressure (BP).

This increase in BP combined with suppressed cortisol (a hormone responsible for managing stress response and alertness) when the individual is under some kind of stress could lead to heart diseases. Repeated interruption of melatonin production during nights can result in sleep disorders leading, especially, to lesser sleep.

At daybreak, the same SCN triggers the production of other hormones like:

- Dopamine, needed for pleasure, alertness, and muscle co-ordination
- Serotonin, which controls our impulses and carbohydrate cravings
- Cortisol, needed for stress response and alertness. Remember caffeine in the morning interferes with the production of cortisol (opposite of what we think!)

- Melatonin, which leads to drowsiness, good sleep, lower body temperature, and lower blood pressure

Multiple research results have shown that the ambient light impacts circadian rhythm through three parameters, which are:

1. Spectrum of the light
2. Intensity of light
3. Time of light getting switched on and off

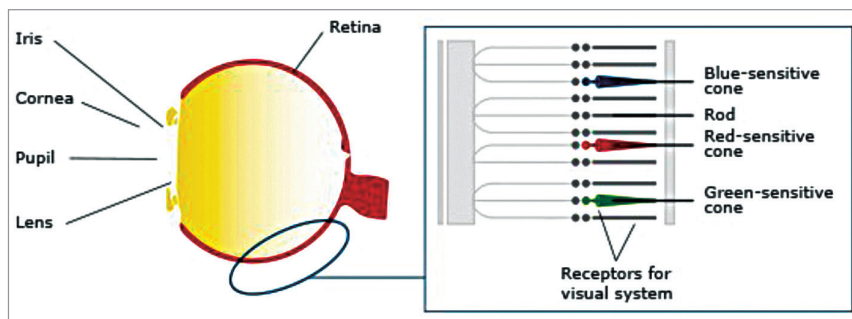


Fig. 2: The three photoreceptors in a human eye

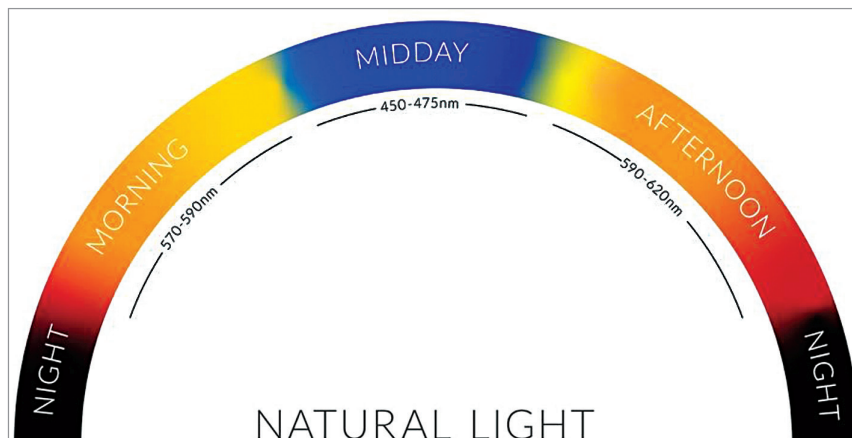


Fig. 3: Light spectrum during different parts of a day

By controlling these three parameters, circadian rhythm can be controlled and tuned for a healthy life. Research has shown that it is important to keep the circadian rhythm in balance, especially of paramedical workers like nurses, patients, and people who work in mines and submarines and so get very little exposure to natural light.

Since circadian rhythms plays a such a vital role, let us briefly look at these three parameters to understand how the light impacts humans and how human-centric lighting can be used for better living.

Spectrum

We all know, white light has seven primary colours. Visible light's wavelength is between 380 and 780nm. Fig. 3 shows how the light spectrum is distributed in a typical day.

Blue wavelength in the natural light is the most biologically active wavelength. Blue light can be produced by selecting a cool white light source. Our eye has red, blue, and green cones to sense colours (see Fig. 2). Interestingly, we do not perceive

all the colours equally. Colours in the region of 555nm wavelength (yellow-green in the visible spectrum) are sensed the brightest.

Ironically, biologically impacting regions are in the wavelength of 480nm and this corresponds to blue light. As we can see, midday light has the maximum blue light, which has a major impact on circadian rhythm. White light is classified into two categories, which are:

1. A white light that has more blue component is called cool white. Fig. 3 shows that midday light has more blue, which leads to lower melatonin secretion and keeps us energised and alert.

2. A white light that has less blue component is called warm white, which we can see in the early mornings and evenings.

Intensity

Second parameter that impacts us is the intensity of the light. Intensity is typically measured in Lux. Studies show that an intensity of 300 to 400 Lux at the eye level, or 1,000 Lux at the work plane level (our working table